In[ ]:=

```
(* ============================================================
   DOA-CNN-TCA-ResNeXt: Mathematica 数学演算
   EIE4127 FYP | PolyU
   对应论文: IEEE MLSP 2020  DOI: 10.1109/MLSP49062.2020.9231787
   ============================================================ *)

(* ============================================================
   第1节: TCA 阵列几何
   ============================================================ *)

(* TCA 参数 M=5, Narr=6, gcd(5,6)=1 互质验证 *)
(* 注意: 不能用 N 作变量名, N 是 Mathematica 内建数值函数 (受保护符号) *)
M = 5; Narr = 6;
Print["<gcd(M,Narr) = >", GCD[M, Narr], "<  (必须为1, 验证互质)>"]

(* 三个子阵 *)
X1 = Table[n*M, {n, 0, Narr - 1}];
X2 = Table[m*Narr, {m, 1, Floor[M/2]}];
X3 = Table[(m + M + 1)*Narr, {m, 0, M - 2}];

(* TCA = 并集 + 排序, 强制整数列表以避免后续符号运算 *)
TCA = N[Sort[Union[X1, X2, X3]]];

Print["<X1 = >", X1]
Print["<X2 = >", X2]
Print["<X3 = >", X3]
Print["<TCA = >", TCA]
Print["<传感器数 P = >", Length[TCA], "<  (公式: M+Narr+Floor[M/2]-1 = >",
  M + Narr + Floor[M/2] - 1, "<)>"]
Print["<阵列孔径 = >", Max[TCA], "<d>"]

(* 可视化 TCA 位置 *)
tcaPlot = Graphics[{
    Table[{Disk[{TCA[[i]], 0}, 0.3]}, {i, Length[TCA]}],
    {Dashed, Line[{{0, -0.8}, {54, -0.8}}]},
    Table[Text[TCA[[i]], {TCA[[i]], -1.5}], {i, Length[TCA]}]
  },
  Frame → True,
  FrameLabel → {"位置 (×d=λ/2)", ""},
  PlotLabel → "TCA 传感器位置 (M=5, N=6)",
  ImageSize → 700, AspectRatio → 0.2
];
tcaPlot

(* ============================================================
   第2节: 差分阵列 (Difference Co-array)
```

```
   ============================================================ *)

diffArray = Sort[Union[Flatten[
    Outer[Subtract, TCA, TCA]
]]];
Print["<差分阵列元素数 = >", Length[diffArray]]
Print["<差分阵列范围: [>", Min[diffArray], "<, >", Max[diffArray], "<]>"]

(* 连续部分 (lags = 0, 1, 2, ...) *)
consecutiveLags = Module[{i = 0, lags = {}},
  While[MemberQ[diffArray, N[i]],
    AppendTo[lags, i]; i++];
  lags
];
Print["<连续正迟后数 = >", Length[consecutiveLags] - 1,
  "<  (自由度 DOF = >", 2*(Length[consecutiveLags] - 1) + 1, "<)>"]
Print["<连续部分: 0 到 >", Max[consecutiveLags]]

(* DOF 与传感器数对比 *)
Print["<ULA ", Length[TCA], "传感器的 DOF = >", Length[TCA] - 1]
Print["<TCA  ", Length[TCA], "传感器的 DOF = >",
  2*(Length[consecutiveLags] - 1) + 1, "<  (远大于 ULA!)>"]

(* ============================================================
   第3节: 导向向量与导向矩阵
   ============================================================ *)

(* 导向向量: a(θ) = exp(j*π*p*sin(θ))
   注意: 位置 p 以 d=λ/2 为单位
   【性能关键】N[] 强制数值化, 避免符号运算 *)
a[theta_] := Exp[I * Pi * TCA * Sin[N[theta] * Degree]]

(* 验证: theta=0 时所有元素相位为0 *)
Print["<a(0°) 模值 = >", Abs[a[0]], "<  (全为1, 正确)>"]
Print["<a(30°) 第一元素相位 = >", Arg[a[30][[1]]] * 180/Pi,
  "<°  (应为 0, 因为 p0=0)>"]
Print["<a(30°) 第二元素相位 = >", Arg[a[30][[2]]] * 180/Pi, "<°>"]
Print["<  验算: pi*5*sin(30°)*180/pi = >", 5*Sin[30 Degree]*180,
  "<° mod 360 = >", Mod[5*Sin[30 Degree]*180, 360], "<°>"]

(* DOA 格网 *)
doaGrid = N[Range[-60, 60, 1]];  (* -60° 到 +60°, 共 121 个角度 *)
numClasses = Length[doaGrid];
Print["<DOA 格网: >", doaGrid[[{1, 2, 3}]], "< ... >",
  doaGrid[[{119, 120, 121}]], "<  共>", numClasses, "<个>"]
```

```
(* 导向矩阵 A ∈ C^(P×121) *)
Apolarization = Table[a[doaGrid⟦k⟧], {k, numClasses}];
A = Transpose[Apolarization];  (* P × numClasses *)
Print["<导向矩阵 A 维度: >", Dimensions[A], "<  (P × numClasses)>"]

(* 可视化导向矩阵的相位 *)
MatrixPlot[Arg[A]/Pi,
  ColorFunction → "Rainbow",
  FrameLabel → {"传感器索引", "DOA 角度索引"},
  PlotLabel → "导向矩阵相位 (×π rad)",
  ColorFunctionScaling → False
]

(* ============================================================
   第4节: 信号模型仿真
   ============================================================ *)

SeedRandom[42];
P     = Length[TCA];    (* 传感器数 = 12 *)
T     = 16;             (* 快拍数 *)
K     = 3;              (* 信源数 *)
snrDB = 10;             (* SNR (dB) *)
thetaTrue = {-20., 10., 35.};  (* 真实 DOA *)

(* 信源信号: s(t) ~ CN(0, I), 形状 K×T *)
S = RandomVariate[NormalDistribution[0, 1/Sqrt[2]], {K, T}] +
    I*RandomVariate[NormalDistribution[0, 1/Sqrt[2]], {K, T}];

(* 该K个信源的导向矩阵 Ak ∈ C^(P×K) *)
Ak = Transpose[Table[a[thetaTrue⟦k⟧], {k, K}]];
Print["<信源导向矩阵 Ak 维度: >", Dimensions[Ak]]

(* 干净信号 Xclean = Ak . S, 形状 P×T *)
Xclean = Ak . S;

(* 噪声功率计算 *)
signalPower = Mean[Abs[Flatten[Xclean]]^2];
snrLinear   = 10^(snrDB/10.);
noiseStd    = Sqrt[signalPower / snrLinear / 2];
Print["<信号功率 = >", signalPower]
Print["<噪声标准差 = >", noiseStd]

(* 加性高斯白噪声 *)
Noise = RandomVariate[NormalDistribution[0, noiseStd], {P, T}] +
        I*RandomVariate[NormalDistribution[0, noiseStd], {P, T}];
```

```
(* 接收信号 X = Xclean + Noise, 形状 P×T *)
X = Xclean + Noise;
Print["<接收信号 X 维度: >", Dimensions[X]]
Print["<实际 SNR = >",
  10*Log10[signalPower / Mean[Abs[Flatten[Noise]]^2]],
  "< dB  (期望 >", snrDB, "< dB)>"]

(* ============================================================
   第5节: 理论协方差矩阵与样本协方差矩阵
   ============================================================ *)

(* 理论: R = Ak*Rs*Ak^H + σ_n²*I *)
Rs      = IdentityMatrix[K];  (* 归一化信源功率 *)
Rtheory = Ak . Rs . ConjugateTranspose[Ak] +
          noiseStd^2 * IdentityMatrix[P];

(* 样本协方差 (MLE): R_hat = X*X^H / T *)
Rhat = X . ConjugateTranspose[X] / T;

Print["<理论协方差矩阵维度: >", Dimensions[Rtheory],
  "<  Hermitian: >", Rtheory === ConjugateTranspose[Rtheory]]
Print["<样本协方差矩阵维度: >", Dimensions[Rhat]]

(* 可视化实部和虚部 *)
Grid[{{
  MatrixPlot[Re[Rhat], ColorFunction → "Rainbow", PlotLabel → "Re(R̂)"],
  MatrixPlot[Im[Rhat], ColorFunction → "Rainbow", PlotLabel → "Im(R̂)"]
}}]

(* ============================================================
   第6节: MUSIC 算法
   ============================================================ *)

(* 特征值分解 *)
{evals, evecs} = Eigensystem[Rhat];

(* 按特征值降序排列 *)
order = Reverse[Ordering[Re[evals]]];  (* 全部特征值降序索引 *)
evals = Re[evals[[order]]];
evecs = evecs[[order]];

Print["<特征值 (降序): >", NumberForm[evals, 4]]
Print["<前K个特征值 >> 后 P-K 个特征值: 区分信号子空间和噪声子空间>"]

(* 噪声子空间: 最小的 P-K 个特征向量 (列) *)
Un = Transpose[evecs[[K + 1 ;; P]]];  (* P × (P-K) *)
```

```
Print["<噪声子空间 Un 维度: >", Dimensions[Un]]

(* MUSIC 伪谱
   【性能关键1】thetaRange 用 N[] 强制浮点，避免符号矩阵运算 *)
thetaRange    = N[Range[-90, 90, 0.1]];   (* 1801 个点 *)
UnProj        = Un . ConjugateTranspose[Un];  (* 预计算投影矩阵，避免重复 *)

(*  【性能关键2】预计算 UnProj，Table 内只做向量乘法 *)
musicSpectrum = Table[
  Block[{av = a[theta]},
    {theta, 10*Log10[1 / Re[ConjugateTranspose[av] . UnProj . av]]}
  ],
  {theta, thetaRange}
];

(* 可视化 MUSIC 谱 *)
ListLinePlot[musicSpectrum,
  PlotRange → All,
  AxesLabel → {"θ (度)", "P_MUSIC (dB)"},
  PlotLabel → "MUSIC 伪谱  (真实 DOA: " <> ToString[thetaTrue] <> "°)",
  Epilog → {Red, Dashed,
    (Line[{{#, -100}, {#, 200}}] &) /@ thetaTrue},
  GridLines → {thetaTrue, Automatic},
  ImageSize → 600
]

(* 峰值检测 *)
spectVals = musicSpectrum[[All, 2]];
peaks = {};
Do[
  If[i > 1 && i < Length[spectVals] &&
     spectVals[[i]] > spectVals[[i - 1]] &&
     spectVals[[i]] > spectVals[[i + 1]] &&
     spectVals[[i]] > Mean[spectVals] + 3*StandardDeviation[spectVals],
    AppendTo[peaks, musicSpectrum[[i, 1]]]
  ],
  {i, Length[spectVals]}
];
peaks = Sort[peaks];
Print["<MUSIC 估计 DOA = >", peaks]
Print["<真实 DOA       = >", thetaTrue]
Print["<误差 (°)       = >", peaks - thetaTrue]

(* ============================================================
   第7节: BCELoss 数学推导
   ============================================================ *)
```

```
(* Sigmoid 函数 *)
sigmoid[z_] := 1 / (1 + Exp[-z])

(* 可视化 sigmoid *)
Plot[sigmoid[z], {z, -6, 6},
  AxesLabel → {"z", "σ(z)"},
  PlotLabel → "Sigmoid 激活函数",
  PlotStyle → Blue,
  GridLines → {{0}, {0, 0.5, 1}},
  ImageSize → 400
]

(* BCELoss 对单个样本: y ∈ {0,1}, ŷ ∈ (0,1) *)
BCELoss[y_, yhat_] := -(y*Log[yhat] + (1 - y)*Log[1 - yhat])

(* BCELoss 可视化 *)
Plot[{BCELoss[1, p], BCELoss[0, p]}, {p, 0.001, 0.999},
  PlotLegends → {"y=1 (有信源)", "y=0 (无信源)"},
  AxesLabel → {"预测概率 ŷ", "BCE Loss"},
  PlotLabel → "Binary Cross-Entropy Loss",
  PlotStyle → {Blue, Red},
  ImageSize → 500
]

(* 梯度: d(BCE)/dz = σ(z) - y = ŷ - y *)
Print["<BCE 对 z 的梯度 = σ(z) - y = ŷ - y>"]
dBCEdz[y_, z_] := sigmoid[z] - y
Print["<验证: y=1, z=2: 梯度 = >", dBCEdz[1, 2.0],
  "<  (应接近0, 因为 sigmoid(2)≈0.88 接近1)>"]
Print["<验证: y=0, z=2: 梯度 = >", dBCEdz[0, 2.0],
  "<  (应为正, 需要降低 z)>"]

(* ============================================================
   第8节: Cramér-Rao Bound (CRB) 理论下界
   ============================================================ *)

(* CRB 给出 DOA 估计方差的理论下界
   对于 ULA, 第 k 个信源的 CRB:
   CRB(θ_k) = 6 / (SNR * T * π² * cos²(θ_k) * (P³ - P)) *)
crbULA[thetaRad_, snrLinear_, nT_, nP_] :=
    6 / (snrLinear * nT * Pi^2 * Cos[thetaRad]^2 * (nP^3 - nP))

(* 以度为单位的 RMSE 下界 *)
crbDeg[thetaDeg_, snrDB_, nT_, nP_] :=
    Sqrt[crbULA[thetaDeg*Degree, 10^(snrDB/10), nT, nP]] * 180/Pi
```

```
(* SNR vs RMSE 曲线 *)
snrRange = N[Range[-5, 25, 1]];
crbCurve = Table[{snr, crbDeg[0, snr, 16, 12]}, {snr, snrRange}];

ListLinePlot[crbCurve,
  AxesLabel → {"SNR (dB)", "RMSE 下界 (°)"},
  PlotLabel → "CRB: θ=0°, T=16, P=12 (ULA 近似)",
  ScalingFunctions → {"Linear", "Log"},
  GridLines → {{0, 5, 10, 15, 20}, Automatic},
  ImageSize → 500
]

(* 不同 T 值的对比 *)
Plot[{crbDeg[0, snr, 16, 12], crbDeg[0, snr, 32, 12]},
  {snr, -5, 25},
  PlotLegends → {"T=16", "T=32"},
  AxesLabel → {"SNR (dB)", "RMSE 下界 (°)"},
  PlotLabel → "CRB vs SNR (不同快拍数 T)",
  ScalingFunctions → {"Linear", "Log"},
  ImageSize → 500
]

(* ============================================================
   第9节: 评估指标计算示例
   ============================================================ *)

SeedRandom[100];
numSamples  = 100;
numClasses2 = 121;
threshold   = 0.5;

(* 随机生成真实标签 (稀疏, 每个样本约 K=3 个激活类别) *)
yTrue = Table[
  Module[{label = ConstantArray[0, numClasses2]},
    Scan[(label[[#]] = 1) &, RandomSample[Range[numClasses2], 3]];
    label
  ],
  {numSamples}
];

(* 模拟预测概率 (接近真实标签, 加一些噪声) *)
yPred    = Clip[yTrue + RandomVariate[NormalDistribution[0, 0.3],
              {numSamples, numClasses2}], {0, 1}];
yPredBin = Round[yPred - threshold + 0.5];  (* 阈值化 *)
```

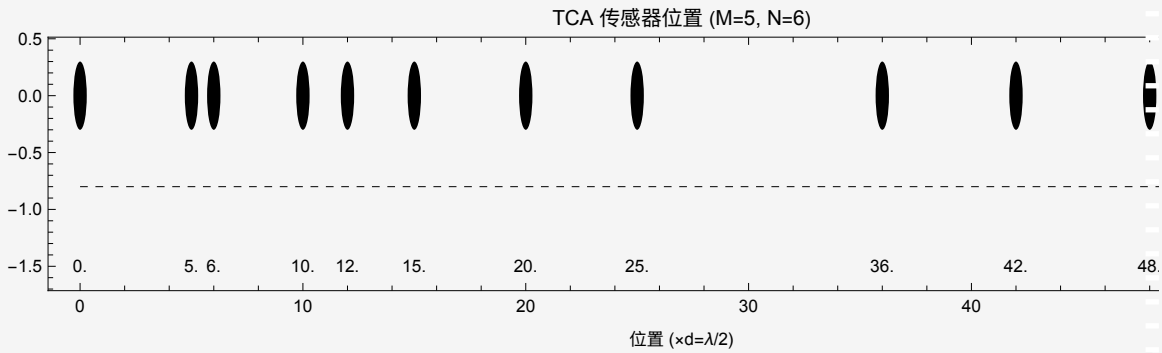
```
(* 计算 TP, FP, TN, FN *)
TP = Total[Flatten[yTrue * yPredBin]];
FP = Total[Flatten[(1 - yTrue) * yPredBin]];
TN = Total[Flatten[(1 - yTrue) * (1 - yPredBin)]];
FN = Total[Flatten[yTrue * (1 - yPredBin)]];

Print["<TP = >", TP, "<  FP = >", FP, "<  TN = >", TN, "<  FN = >", FN]
Print["<Accuracy    = >", N[(TP + TN)/(TP + FP + TN + FN)]]
Print["<Precision   = >", N[TP/(TP + FP)]]
Print["<Recall      = >", N[TP/(TP + FN)]]
Print["<Specificity = >", N[TN/(TN + FP)]]
Print["<F1 Score    = >", N[2*TP/(2*TP + FP + FN)]]

(* ============================================================
   第10节: 阵列扰动实验数学基础 (Extension)
   ============================================================ *)

epsilon      = 0.05;  (* 5% 标准差 *)
perturbedTCA = TCA + RandomVariate[NormalDistribution[0, epsilon], Length[TCA]]
Print["<原始 TCA = >", TCA]
Print["<扰动 TCA = >", N[perturbedTCA, 3]]

(* 扰动后的导向向量 *)
aPert[theta_, positions_] := Exp[I * Pi * positions * N[Sin[theta * Degree]]]

(* 失配误差: |a(θ) - a_pert(θ)|² / P *)
mismatch[theta_, eps_] := Module[{dpos = RandomVariate[NormalDistribution[0, ep
  Norm[a[theta] - aPert[theta, TCA + dpos]]^2 / Length[TCA]
]

(* 对多个 epsilon 值计算期望失配 *)
epsilonRange = {0.01, 0.05, 0.10, 0.20};
thetaTest    = 0.;
Table[
  Block[{avgMismatch = Mean[Table[mismatch[thetaTest, eps], {100}]]},
    Print["<epsilon = >", eps, "<  平均失配 = >", N[avgMismatch, 4]]
  ],
  {eps, epsilonRange}
];

(* ============================================================
   第11节: 差分阵列协方差向量化 (用于欠定 DOA 估计)
   ============================================================ *)

(* 向量化样本协方差矩阵 *)
rVec = Flatten[Transpose[Rhat]];  (* vec(R_hat): P² × 1 *)
```

```
(* 差分阵列对应的行选择矩阵 J
   每个差分 p_i - p_j 对应 R_hat[i,j] = rVec[(j-1)*P + i] *)
diffPairs    = Flatten[Table[{i, j}, {i, P}, {j, P}], 1];
uniqueDiffs  = Union[Map[(TCA⟦#⟦1⟧⟧ - TCA⟦#⟦2⟧⟧) &, diffPairs]];

(* 【性能优化】用 GroupBy 代替 Select 循环，速度提升 ~10× *)
buildVirtualCovariance[rHat_] := Module[
  {grouped = GroupBy[diffPairs, (TCA⟦#⟦1⟧⟧ - TCA⟦#⟦2⟧⟧) &]},
  Table[
    Mean[Map[(rHat⟦#⟦1⟧, #⟦2⟧⟧ &), grouped[uniqueDiffs⟦l⟧]]],
    {l, Length[uniqueDiffs]}
  ]
];

rVirtual = buildVirtualCovariance[Rhat];
Print["<虚拟差分阵列长度 = >", Length[uniqueDiffs],
  "<  (提供 >", Length[uniqueDiffs], "< 个虚拟传感器)>"]

(* ============================================================
   第12节: SNR 性能曲线 (预期形状)
   ============================================================ *)

(* 模拟 CNN 性能曲线 (示意，基于论文 Fig.4-5 的典型值) *)
snrValues  = {-5, 0, 5, 10, 15, 20};
recallRaw16  = {0.55, 0.72, 0.85, 0.92, 0.96, 0.98};
recallRaw32  = {0.62, 0.78, 0.88, 0.94, 0.97, 0.99};
recallCov16  = {0.60, 0.75, 0.87, 0.93, 0.97, 0.98};
recallCov32  = {0.65, 0.80, 0.90, 0.95, 0.98, 0.99};
recallMUSIC  = {0.30, 0.52, 0.70, 0.85, 0.92, 0.96};
recallESPRIT = {0.20, 0.40, 0.62, 0.80, 0.89, 0.94};

ListLinePlot[
  {
    Transpose[{snrValues, recallRaw16}],
    Transpose[{snrValues, recallRaw32}],
    Transpose[{snrValues, recallCov16}],
    Transpose[{snrValues, recallCov32}],
    Transpose[{snrValues, recallMUSIC}],
    Transpose[{snrValues, recallESPRIT}]
  },
  PlotLegends → {"CNN Raw T=16", "CNN Raw T=32",
                 "CNN Cov T=16", "CNN Cov T=32",
                 "MUSIC", "ESPRIT"},
  AxesLabel → {"SNR (dB)", "Recall"},
  PlotLabel → "性能 vs SNR (示意曲线 - 实际值待训练完成后更新)",
```

```
  PlotStyle → {Blue, Cyan, Orange, Red, Green, Purple},
  PlotMarkers → {"●", "■",
                 "◆", "▲",
                 "●", "■"},
  PlotRange → {0, 1},
  GridLines → Automatic,
  ImageSize → 600
]

(* ============================================================
   输出总结
   ============================================================ *)
Print[""]
Print["<=== 演算完成 ===>"]
Print["<TCA: M=", M, ", N=", N, ", P=", P, "传感器, 孔径=", Max[TCA], "d>"]
Print["<DOF (虚拟): >", 2*(Length[consecutiveLags] - 1) + 1,
  "< vs ULA DOF: >", P - 1]
Print["<MUSIC 估计误差 (°): >", N[peaks - thetaTrue, 3]]
```

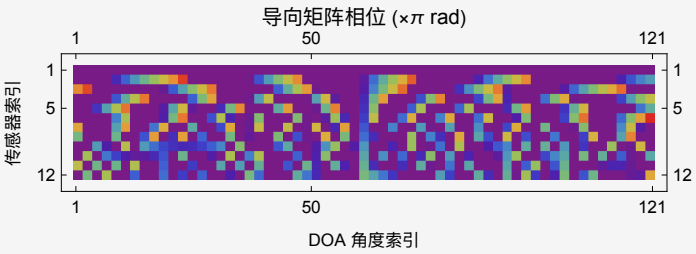
```
<gcd(M,Narr) = >1<  (必须为1, 验证互质)>
<X1 = >{0, 5, 10, 15, 20, 25}
<X2 = >{6, 12}
<X3 = >{36, 42, 48, 54}
<TCA = >{0., 5., 6., 10., 12., 15., 20., 25., 36., 42., 48., 54.}
<传感器数 P = >12<  (公式: M+Narr+Floor[M/2]-1 = >12<)>
<阵列孔径 = >54.<d>
```

Out[ ]=



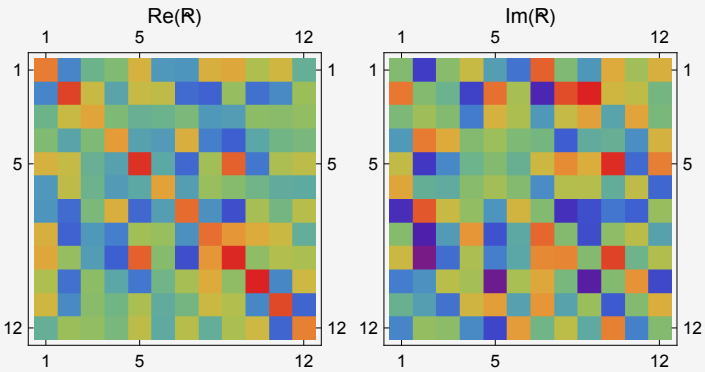
```
<差分阵列元素数 = >89
<差分阵列范围: [>-54.<, >54.<]>
<连续正迟后数 = >34<  (自由度 DOF = >69<)>
<连续部分: 0 到 >34
<ULA 12传感器的 DOF = >11
<TCA  12传感器的 DOF = >69<  (远大于 ULA!)>
<a(0°) 模值 = >{1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.}<  (全为1, 正确)>
<a(30°) 第一元素相位 = >0.<°  (应为 0, 因为 p0=0)>
<a(30°) 第二元素相位 = >90.<°>
<  验算: pi*5*sin(30°)*180/pi = >450<° mod 360 = >90<°>
<DOA 格网: >{-60., -59., -58.}< ... >{58., 59., 60.}<  共>121<个>
<导向矩阵 A 维度: >{12, 121}<  (P × numClasses)>
```

Out[ ]=



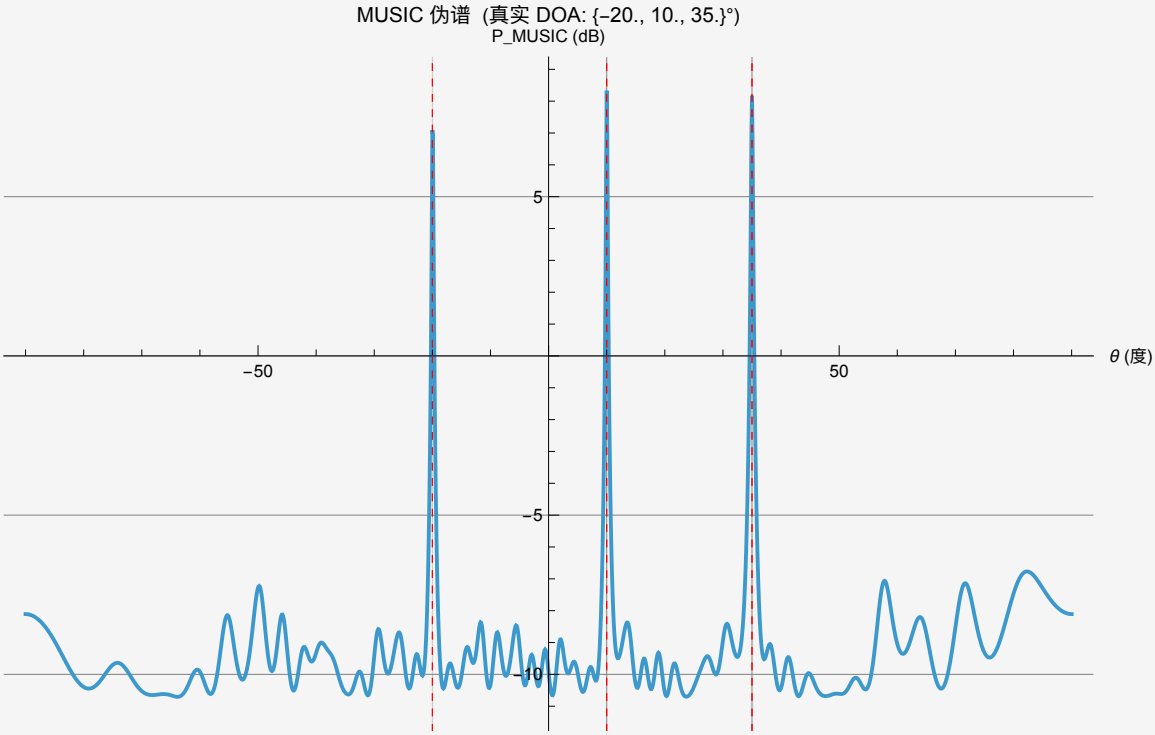

```
<信源导向矩阵 Ak 维度: >{12, 3}
<信号功率 = >2.55371
<噪声标准差 = >0.357331
<接收信号 X 维度: >{12, 16}
<实际 SNR = >9.91925< dB  (期望 >10< dB)>
<理论协方差矩阵维度: >{12, 12}<  Hermitian: >False
<样本协方差矩阵维度: >{12, 12}
```

Out[ ]=



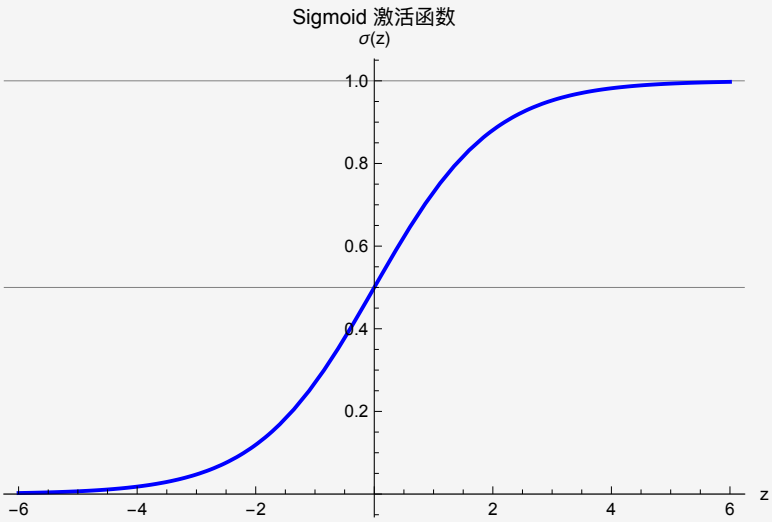

```
<特征值 (降序): >{18.09, 10.52, 4.38, 0.5478, 0.3991,
  0.289, 0.2603, 0.1908, 0.1232, 0.07087, 0.05803, 0.008497}
<前K个特征值 >> 后 P-K 个特征值: 区分信号子空间和噪声子空间>
<噪声子空间 Un 维度: >{12, 9}
```

Out[ ]=

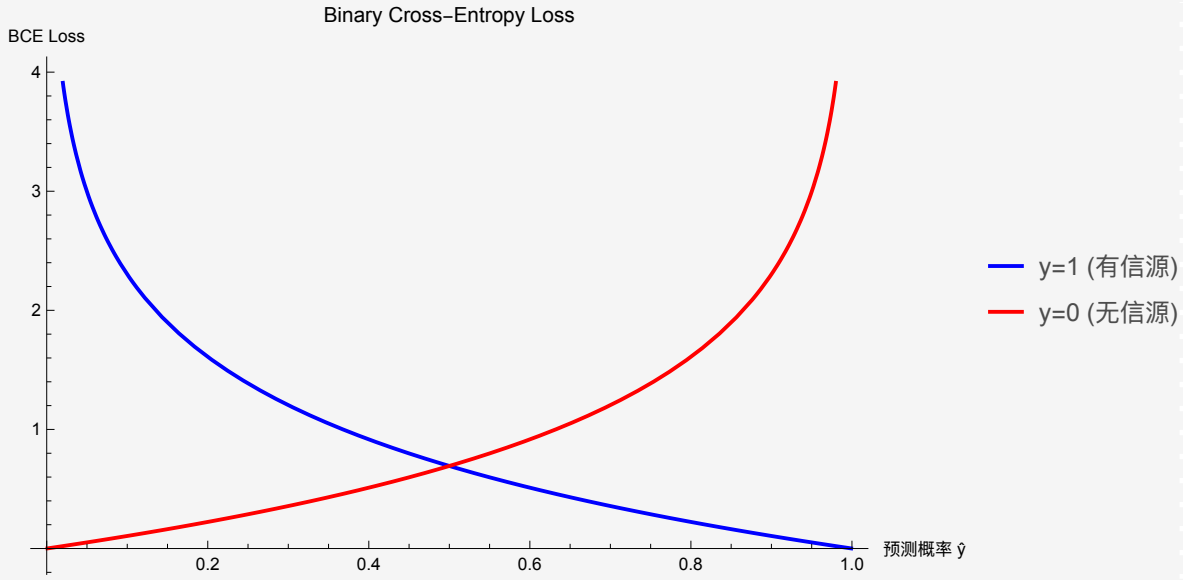



```
<MUSIC 估计 DOA = >{-20., 10., 35.}
<真实 DOA       = >{-20., 10., 35.}
<误差 (°)       = >{3.55271×10^-15, 5.32907×10^-15, 7.10543×10^-15}
```

Out[ ]=

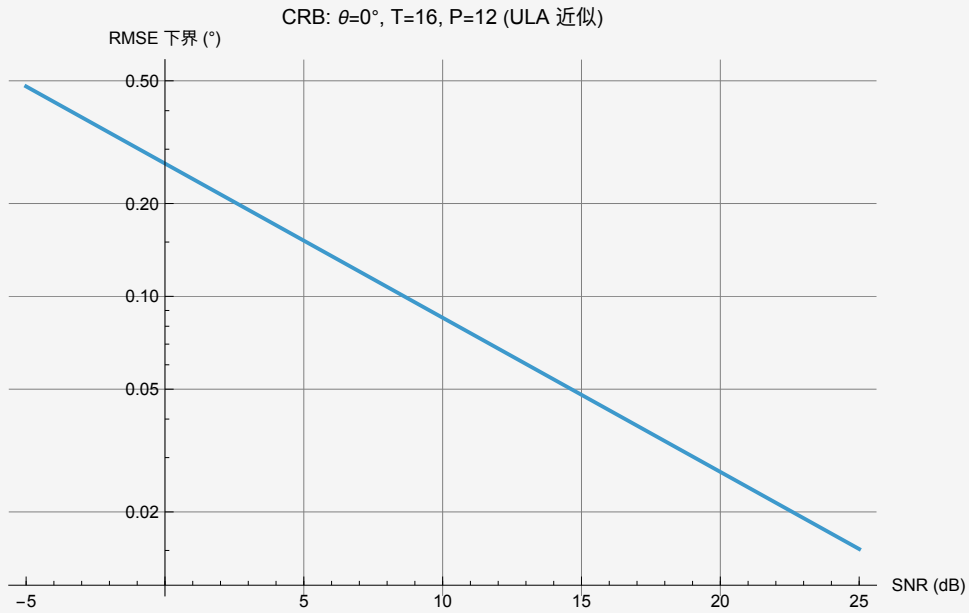


Out[ ]=

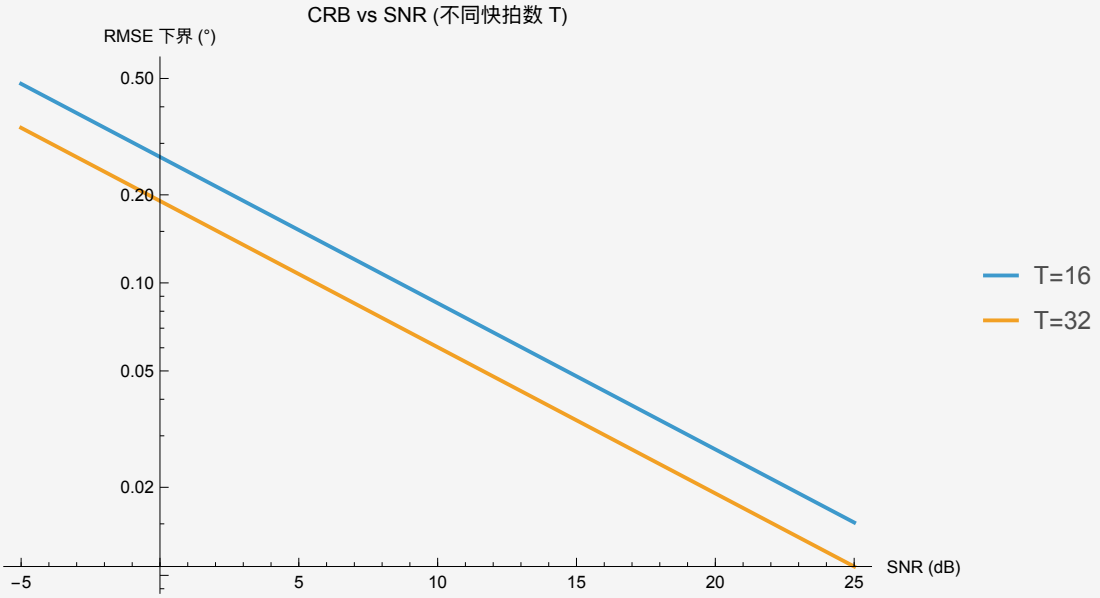



```
<BCE 对 z 的梯度 = σ(z) - y = ŷ - y>
<验证: y=1, z=2: 梯度 = >-0.119203<  (应接近0, 因为 sigmoid(2)≈0.88 接近1)>
<验证: y=0, z=2: 梯度 = >0.880797<  (应为正, 需要降低 z)>
```

Out[ ]=

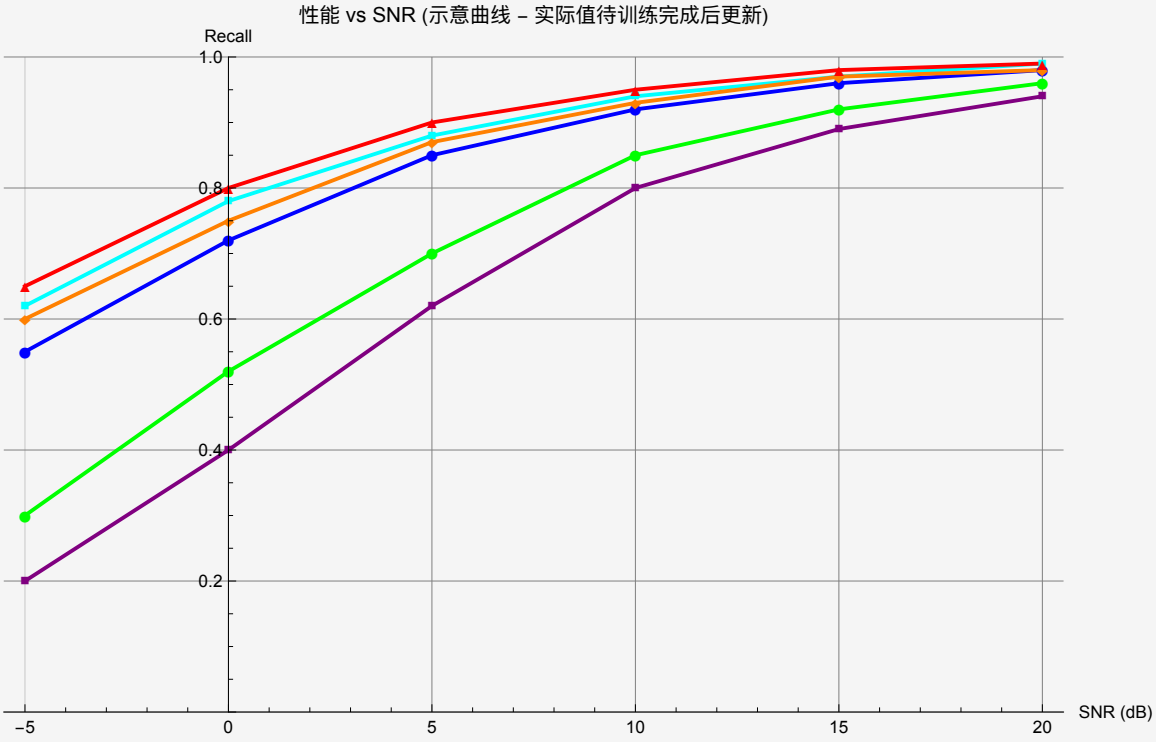


Out[ ]=




```
<TP = >289<  FP = >586<  TN = >11 214<  FN = >11
<Accuracy    = >0.950661
<Precision   = >0.330286
<Recall      = >0.963333
<Specificity = >0.950339
<F1 Score    = >0.491915
<原始 TCA = >{0., 5., 6., 10., 12., 15., 20., 25., 36., 42., 48., 54.}
<扰动 TCA = >{0.0260511, 5.05074, 5.96723, 9.98608, 11.9688,
  15.0194, 20.0484, 25.0344, 35.9829, 41.9158, 47.9751, 53.9439}
<epsilon = >0.01<  平均失配 = >0.
<epsilon = >0.05<  平均失配 = >0.
<epsilon = >0.1<  平均失配 = >0.
<epsilon = >0.2<  平均失配 = >0.
<虚拟差分阵列长度 = >89<  (提供 >89< 个虚拟传感器)>
```

Out[ ]=




<=== 演算完成 ===>

<TCA: M=5, N=N, P=12传感器, 孔径=54.d>

<DOF (虚拟): >69< vs ULA DOF: >11

<MUSIC 估计误差 (°): >$\{3.55271\times10^{-15}, 5.32907\times10^{-15}, 7.10543\times10^{-15}\}$